

5.5.1:

**Solution:** We can start by finding the exact solution using separation of variables:

$$\begin{aligned} x' = 2t(1+x) &\Rightarrow \int \frac{1}{1+x} dx = \int 2t dt \\ &\Rightarrow \ln |1+x| = t^2 + C \\ &\Rightarrow x = Ce^{t^2} - 1 \\ &\Rightarrow 0 = C - 1 \Rightarrow C = 1 \\ &\Rightarrow \boxed{x(t) = e^{t^2} - 1} \end{aligned}$$

Now, using Picard Iterations we can find  $X_1$  through  $X_4$  given  $X_0 = 0$ :

$$\begin{aligned} X_0 = 0 &\Rightarrow X_1 = \int_0^t 2s(1+0) ds = \int_0^t 2s ds = t^2 \\ &\Rightarrow X_2 = \int_0^t 2s(1+X_1(s)) ds = \int_0^t 2s(1+s^2) ds = t^2 + \frac{t^4}{2} \\ &\Rightarrow X_3 = \int_0^t 2s(1+X_2(s)) ds = \int_0^t 2s \left(1+s^2+\frac{s^4}{2}\right) ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6} \\ &\Rightarrow X_4 = \int_0^t 2s(1+X_3(s)) ds = \int_0^t 2s \left(1+s^2+\frac{s^4}{2}+\frac{s^6}{6}\right) ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6} + \frac{t^8}{24} \end{aligned}$$

Looking at the Taylor series of the exact solution,

$$e^{t^2} - 1 = \sum_{k=0}^{\infty} \frac{t^{2k}}{k!} = t^2 + \frac{t^4}{2} + \frac{t^6}{6} + \frac{t^8}{24} + \dots$$

it is clear that the  $n$ -th Picard iteration is equal to the  $n$ -th partial sum of the Taylor series.

5.5.2:

**Solution:** Let  $X_0 = \vec{x}_0 = \begin{pmatrix} a \\ b \end{pmatrix}$ . Then

$$\begin{aligned} X_1(t) &= \vec{x}_0 + \int_s^t A(\tau)x_0 d\tau \\ &= \begin{pmatrix} a \\ b \end{pmatrix} + \int_s^t \begin{pmatrix} b \\ a\tau^{-2} - b\tau^{-1} \end{pmatrix} d\tau \\ &= \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} b(t-s) \\ a\left(\frac{1}{s} - \frac{1}{t}\right) - b \ln\left(\frac{t}{s}\right) \end{pmatrix} \\ &= \begin{pmatrix} a + b(t-s) \\ b + a\left(\frac{1}{s} - \frac{1}{t}\right) - b \ln\left(\frac{t}{s}\right) \end{pmatrix}. \end{aligned}$$

It follows that

$$\begin{aligned} X_2(t) &= \vec{x}_0 + \int_s^t A(\tau) X_1(\tau) d\tau \\ &= \int_s^t \left( b + a \left( \frac{1}{s} - \frac{1}{\tau} \right) - b \ln \left( \frac{\tau}{s} \right) \right) d\tau \\ &= \begin{pmatrix} a + 2b(t-s) + \frac{a}{s}(t-s) - (a+bt) \ln \left( \frac{t}{s} \right) \\ b + 2a \left( \frac{1}{s} - \frac{1}{t} \right) - \frac{a}{s} \ln \left( \frac{t}{s} \right) + b \left( 1 - \frac{s}{t} \right) + \frac{b}{2} \ln \left( \frac{t}{s} \right)^2 \end{pmatrix}. \end{aligned}$$

So,

$$\begin{aligned} X_3 &= \vec{x}_0 + \int_s^t A(\tau) X_2(\tau) d\tau \\ &= \begin{pmatrix} \left( \frac{3t}{s} - 2 + \left( 2 + \frac{t}{s} \right) \ln \left( \frac{s}{t} \right) \right) a + \left( (t-s) \left( 1 + \ln \left( \frac{s}{t} \right) \right) + \frac{t}{2} \ln \left( \frac{s}{t} \right)^2 \right) b \\ \left( \frac{(t-s)(1+\ln(\frac{s}{t}))}{st} + \frac{1}{2s} \ln \left( \frac{s}{t} \right)^2 \right) a + \left( \frac{3s}{t} - 2 - 2 \ln \left( \frac{s}{t} \right) - \frac{1}{2} \ln \left( \frac{s}{t} \right)^2 + \frac{1}{6} \ln \left( \frac{s}{t} \right)^3 \right) b \end{pmatrix}. \end{aligned}$$

It is clear that the Picard solutions converge to the exact solution (for  $t \neq 0$ ). Formally,

$$X_n(t) = \Phi_{t,s}^{(n)} \vec{x}_0 \rightarrow \Phi_{t,s} \vec{x}_0 \quad \text{as } n \rightarrow \infty$$

where  $X_n(t) = \Phi_{t,s}^{(n)} \vec{x}_0$ .

### 5.5.3:

**Solution:** Start by defining  $x_1 = x$  and  $x_2 = x'$  so

$$x'_1 = x_2 \quad \text{and} \quad x'_2 = x'' = \frac{2x}{t^2} + \frac{3}{t^3} = \frac{2x_1}{t^2} + \frac{3}{t^3}.$$

Thus,

$$\vec{x}'(t) = \begin{pmatrix} 0 & 1 \\ \frac{2}{t^2} & 0 \end{pmatrix} \vec{x}(t) + \begin{pmatrix} 0 \\ \frac{3}{t^3} \end{pmatrix}, \quad \vec{x}(1) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

where

$$A(t) = \begin{pmatrix} 0 & 1 \\ \frac{2}{t^2} & 0 \end{pmatrix} \quad \text{and} \quad \vec{b}(t) = \begin{pmatrix} 0 \\ \frac{3}{t^3} \end{pmatrix}, \quad t_0 = 1.$$

Next, we solve the homogenous system

$$\vec{x}' = A(t)\vec{x} \Rightarrow t^2 x'' - 2x = 0.$$

This has solutions  $x = t^2$  and  $x = t^{-1}$  so

$$M(t) = \begin{pmatrix} t^2 & t^{-1} \\ 2t & -t^{-2} \end{pmatrix}.$$

Considering  $\Phi_{t,s} = M(t)M^{-1}(s)$  at  $s = t_0 = 1$  we have

$$M(1) = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \Rightarrow M(1)^{-1} = -\frac{1}{3} \begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1/3 & 1/3 \\ 2/3 & -1/3 \end{pmatrix}$$

so

$$\Phi_{t,1} = \begin{pmatrix} t^2 & t^{-1} \\ 2t & -t^{-2} \end{pmatrix} \begin{pmatrix} 1/3 & 1/3 \\ 2/3 & -1/3 \end{pmatrix} = \begin{pmatrix} \frac{t^2}{3} + \frac{2}{3t} & \frac{t^2}{3} - \frac{1}{3t} \\ \frac{2t}{3} + \frac{1}{3t^2} & \frac{2t}{3} - \frac{1}{3t^2} \end{pmatrix}.$$

Note that we also have

$$M(s)^{-1} = -\frac{1}{3} \begin{pmatrix} -s^{-2} & -s^{-1} \\ -2s & s^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{3s^2} & \frac{1}{3s} \\ \frac{2s}{3} & -\frac{s}{3} \end{pmatrix}.$$

So,

$$\vec{x}(t) = \Phi_{t,t_0} \vec{x}_0 + \int_{t_0}^t \Phi_{t,s} \vec{b}(s) ds = \Phi_{t,1} \vec{x}_0 + \int_1^t \begin{pmatrix} t^2 s^{-4} - t^{-1} s^{-1} \\ 2ts^{-4} + t^{-2} s^{-1} \end{pmatrix} ds.$$

Since  $\vec{x}(t) = \begin{pmatrix} x \\ x' \end{pmatrix}$ , we only need to integrate the first row to find  $x$ :

$$\begin{aligned} x(t) &= \left( \frac{t^2}{3} + \frac{2}{3t} + \frac{2t^2}{3} - \frac{2}{3t} \right) + \int_1^t t^2 s^{-4} - t^{-1} s^{-1} ds \\ &= t^2 + \left( \frac{t^2}{3} - \frac{1}{2t} - \frac{\ln t}{t} \right) \\ &= \boxed{\frac{4t^2}{3} - \frac{1}{3t} - \frac{\ln t}{t}} \end{aligned}$$

**5.5.7:**

**Solution:** Start by defining  $X(t) = \begin{pmatrix} x \\ x' \end{pmatrix}$ . Then, we have

$$X'(t) = AX(t) + \vec{b}(t)$$

with

$$A = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix} \quad \text{and} \quad \vec{b}(t) = \begin{pmatrix} 0 \\ e^t \end{pmatrix}.$$

We start by solving the homogenous system  $X' = AX$  which is equivalent to

$$x'' - x' - 6x = 0.$$

This has solutions  $x = e^{3t}$  and  $x = e^{-2t}$ . So, the fundamental matrix is

$$M(t) = \begin{pmatrix} e^{3t} & e^{-2t} \\ 3e^{3t} & -2e^{-2t} \end{pmatrix}.$$

We have that  $\Phi_{t,s} = M(t)M(s)^{-1}$  with

$$M(s)^{-1} = \frac{1}{5} \begin{pmatrix} 2e^{-3s} & e^{-3s} \\ 3e^{2s} & -e^{2s} \end{pmatrix}.$$

Then,

$$\begin{aligned} \Phi_{t,s} &= \frac{1}{5} \begin{pmatrix} e^{3t} & e^{-2t} \\ 3e^{3t} & -2e^{-2t} \end{pmatrix} \begin{pmatrix} 2e^{-3s} & e^{-3s} \\ 3e^{2s} & -e^{2s} \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} 2e^{3(t-s)} + 3e^{-2(t-s)} & e^{3(t-s)} - e^{-2(t-s)} \\ 6e^{3(t-s)} - 6e^{-2(t-s)} & 3e^{3(t-s)} + 2e^{-2(t-s)} \end{pmatrix}. \end{aligned}$$

Since

$$X(t) = \Phi_{t,s} X(s) + \int_s^t \Phi_{t,\tau} \begin{pmatrix} 0 \\ e^\tau \end{pmatrix} d\tau$$

but recall that

$$X(t) = \begin{pmatrix} x \\ x' \end{pmatrix}$$

so we can solve for  $x$  using just the first row. So, the particular solution is

$$\begin{aligned} x_p &= \int_s^t \Phi_{t,\tau} \begin{pmatrix} 0 \\ e^\tau \end{pmatrix} d\tau = \frac{1}{5} \int_s^t (e^{3(t-\tau)} - e^{-2(t-\tau)})e^\tau d\tau \\ &= \frac{1}{5} \left( \frac{e^{3t}}{2} (e^{-2s} - e^{-2t}) - \frac{e^{-2t}}{3} (e^{3t} - e^{3s}) \right) \\ &= \frac{e^{3t-2s}}{10} + \frac{e^{3s-2t}}{15} - \frac{e^t}{6}. \end{aligned}$$

Since the first two terms are homogenous, they can be absorbed into constants, leaving

$$x_p(t) = -\frac{e^t}{6}$$

so

$$x(t) = C_1 e^{3t} + C_2 e^{-2t} - \frac{1}{6} e^t$$